

OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

¹Ayush Debnath, ^{3,4}Ruelia Saha, ²Sudip Misra

¹Dept. of Mechanical Engg., ²Dept. of Computer Sci. and Engg., Indian Institute of Technology Kharagpur, India 721302

³KTH Royal Inst. of Technology, Sweden ⁴SRM University-AP, India

Emails: {ayush.d@kgpian.iitkgp.ac.in, ruelia.saha.rs@gmail.com, sudipm@iitkgp.ac.in}

Abstract—Simultaneous assessment of medical imaging and patient records is often required in clinical diagnosis. However, standard machine learning algorithms cannot analyze these several sorts of data together. Meanwhile, compliance with the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) can constrain centralized aggregation of sensitive patient data. This leaves a crucial void of secure fusion of visual and textual context across distant networks. Thus, we present OmniMed-FL, a controlled systems study of multimodal federated learning for five-class clinical condition classification (Normal, Pneumonia, COVID-19, Pleural Effusion, Cardiomegaly). Our controlled corpus pairs 3,000 public chest radiographs, including 600 non-synthetic COVID-19 images, with 3,000 class-conditioned synthetic notes matched by class rather than patient. The framework benchmarks eight fusion strategies, three initializations, four missing-text imputation rules, and matched federated baselines under non-IID Dirichlet partitioning across 3 to 20 hospital clients. As all notes are synthetic and pairing is not patient-level, these are descriptive proxy comparisons, not estimates of diagnostic performance or deployment readiness. Within those limits, at $K=5$ and severe skew ($\alpha=0.1$), local-only training achieves a macro-F1 score of 0.297, FedAvg achieves 0.662 ± 0.074 , FedProx 0.737 ± 0.085 , a matched FedMME-style one-shot ensemble 0.647 ± 0.080 , and our SCAFFOLD–AdamW adaptation 0.070 ± 0.015 ; the 0.075 FedProx–FedAvg gap lies inside the wider two-seed standard deviation. Over a 4×3 grid, label skew costs up to 0.27 F1 whereas a near-sevenfold client increase costs at most 0.10, while bidirectional volume grows linearly to 183.5 GiB at $K=20$. A pooled diagnostic screen separates fusion operators by 0.176 F1, well above its 0.005 median seed standard deviation, but a matched single-seed federated branch audit scores text/image/concatenation at 0.880/0.768/0.830. The anti-collapse stack does not improve F1: at $\alpha=0.1$, removing both components raises it from 0.683 to 0.758 without eliminating predicted-class coverage.

Index Terms—Multimodal Federated Learning, Vision–Language Models, Clinical Condition Classification, E-Health, Non-Identically Distributed Data, Communication Cost, Retrieval-Augmented Generation

I. INTRODUCTION

CLINICAL condition classification, which covers distinguishing normal presentations from acute infections, fluid accumulation, and structural cardiac pathology, is central to timely patient intervention. Automated systems typically operate on a single modality. For example, vision models trained on chest radiographs [1] do not capture the full diagnostic picture. Image-only models miss contextual cues such as BNP elevation or viral contact history, while text-only models miss radiological

patterns such as bilateral ground-glass opacity or pleural fluid meniscus.

The healthcare setting complicates deployment further because patient data is distributed across hospitals, clinics, and diagnostic centers. Federated learning (FL) addresses these constraints by transmitting only model updates [2], although keeping raw records at home is a data-locality property, not a privacy proof. The cited clinical FL systems are single-modality [3], [4], discarding the notes clinicians record alongside imaging. The aforementioned lacunae is addressed by OmniMed-FL, so hospitals can collaborate on multimodal clinical models without exposing patient data. We measure four properties of such a system, namely how far label skew degrades accuracy, the point at which a five-class head collapses onto its majority class, which fusion operator earns the compute it consumes, and what a multimodal model costs to move. Each is measured on a controlled proxy benchmark whose notes are synthetic and paired by class, so what we characterize is the learning system rather than clinical diagnosis. The v3 rerun therefore focuses on comparative systems questions: aggregation under skew, fusion sensitivity, initialization, missing text, and communication cost.

A. Contribution

The main contributions of this paper are as follows:

- We provide matched local-only, FedAvg, FedProx, SCAFFOLD–AdamW, and FedMME-style controls on a fully public-radiograph proxy corpus, separating the benefit of federation from differences in data, model, and local budget.
- We build a controlled five-class corpus whose notes mix wording across classes, so the label cannot be read off a single keyword, and we test how much of the label a model can still recover from wording alone.
- We quantify skew, initialization, fusion, anti-collapse, runtime, memory, and communication. We retain negative results: the anti-collapse components hurt F1 under severe skew, and the matched text-only branch beats concatenation.

II. RELATED WORK

A. Medical Image Classification

Early automated diagnostics treated the chest radiograph as a pure computer-vision problem. CheXNet and CheXpert set

centralized CNN baselines [1], [5], and Vision Transformers later established an attention-based alternative for general image classification [6]. All of them pool data on one server, and none reads the note a clinician wrote next to the image.

B. Federated Clinical Learning

Privacy regulation moved the clinical community toward FL. FedAvg established decentralized training [2], and later work added multi-institutional collaboration without shared patient records [3], multinational COVID-19 CT validation [4], federated weakly supervised medical-image segmentation [7], and communication-efficient health monitoring [8]. FedProx and SCAFFOLD address the client heterogeneity that degrades naive averaging [9]–[11]. The clinical deployments discussed above remain unimodal; therefore, their behavior after incorporating a second modality remains unexplored.

C. Vision–Language Models in Healthcare

Centralized vision–language models advanced in parallel. BiomedCLIP, LLaVA-Med, and MedCLIP jointly model biomedical visual and textual information using paired, instruction-derived, or deliberately unpaired corpora, respectively [12]–[14], but they train on centrally assembled corpora. Multimodal federated learning is more recent. FedMME combines one-shot client models by voting, and P-FIN fills in missing features with calibrated uncertainty [15], [16], each on its own data and budget, while cross-modal federated medical imaging is also appearing [17]. Neither line reports how much bandwidth and memory a multimodal federated model costs.

Synthesis. The landscape above offers either the multimodal accuracy of centralized vision–language models or the data locality of federated learning, but not both. OmniMed-FL bridges that gap, embedding text–image fusion inside a federated loop that leaves every record at the hospital that holds it.

III. SYSTEM DESIGN

A. Architecture

Figure 1 illustrates the pipeline of the proposed scheme. We tokenize notes with WordPiece at a 128-token cap, resize radiographs to 224×224 , and normalize them against ImageNet statistics. The DistilBERT [18] [CLS] vector is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$, and mean-pooled ViT tokens give $\mathbf{h}_v \in \mathbb{R}^{512}$. Eight fusion rules then combine the text and image vectors. Three use no attention (projected concatenation, dual-projection concatenation, and gated residual fusion) and five do (residual cross-attention, its non-residual 384-d variant, a one-layer decoder A, a two-query/two-layer decoder B, and a two-token Transformer encoder). All eight feed the same two-layer, five-class multilayer perceptron (MLP) and share the data, partition, optimizer, and local budget. The eight rules test whether the attention-free ones score as well as the attention-based ones when both get the same training on hardware one hospital can afford.

B. Federated Learning Setup

Let K clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let θ collect the trainable parameters on the fusion path: both encoders, the projection layers, the fusion operator, and the classification head. We minimize the sample-weighted empirical risk

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad (1)$$

$$F_k(\theta) = \frac{1}{n_k} \sum_{(x,z,y) \in \mathcal{D}_k} \ell(f_{\theta}(x,z), y).$$

where x is a radiograph, z its class-paired note, y the class label, and f_{θ} the fused five-class classifier. Shards never leave a client, and the server sees only the trainable tensors listed above. For mini-batch $\mathcal{B}$, the loss we actually implement is

$$\ell_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (1 - p_{i,y_i})^{\gamma} \text{CE}_{\epsilon,i} + \lambda \phi(h) + 5[q - 0.9]_+, \quad (2)$$

with p_{i,y_i} the true-class probability, $\text{CE}_{\epsilon,i}$ label-smoothed cross-entropy, $\gamma=2$, $\epsilon=0.05$, $C=5$, $h = H(\bar{p})/\log C$, $q = |\mathcal{B}|^{-1} \sum_i \max_c p_{ic}$, $\phi(h) = (1-h)[1 - \frac{8}{3}[h - 0.7]_+]_+$, and $[u]_+ = \max(u, 0)$. The entropy-diversity term grows whenever a batch’s predictions concentrate, and we attenuate it above normalized entropy 0.7 so it stops pushing once the head is healthy. Setting $\lambda=0$ removes that term alone, while the confidence penalty $5[q-0.9]_+$ stays fixed everywhere. The class-balanced sampler and the entropy-diversity term are the two anti-collapse components we ablate in Section IV. Predicted-class diversity is the fraction of the five labels that ever appear as a validation-set argmax.

Each round, the server broadcasts $\theta^{(t)}$, every eligible client runs three local AdamW epochs on class-balanced mini-batches under eq. (2), clips the gradient norm at one, and returns its trainable tensors for the sample-weighted update in eq. (1). Operational runs start from public DistilBERT and ViT-Base/16 weights with random task heads. We fine-tune both encoders and keep unused unimodal heads and the inactive CNN fallback out of optimization and communication.

C. Controlled Proxy Corpus

We built a balanced 3,000-image proxy corpus at 600 examples per class. Every image is a public radiograph: sources include the Kermany pneumonia collection [19], the NIH ChestX-ray dataset [20], and the COVID-19 Radiography Database [21]. The controlled-v3 loader disables synthetic image generation and raises on any source shortfall; its metadata records 3,000 public radiographs and no synthetic COVID-19 images. Each class is drawn from a collection that labels it, which leaves the image axis partly confounded with source collection. Images are resized to 224×224 , ImageNet-normalized, shuffled deterministically, and split 80/20 into 2,400 training and 600 validation examples.

Clinical text data. We use no patient electronic health record (EHR) text, and the notes come from a template engine. It pairs

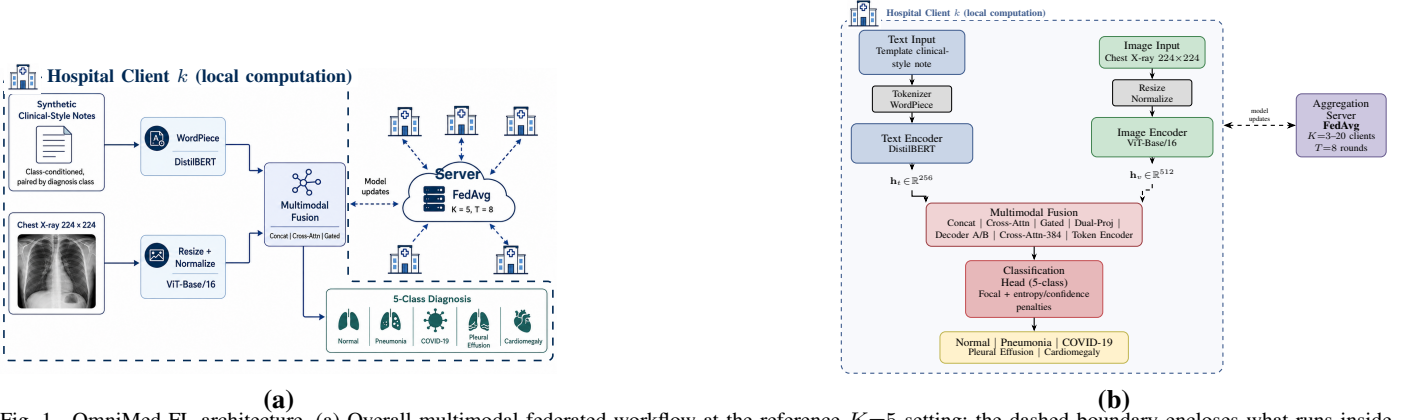


Fig. 1. OmniMed-FL architecture. (a) Overall multimodal federated workflow at the reference $K=5$ setting; the dashed boundary encloses what runs inside a single hospital client, and only model updates cross it to the FedAvg cloud shared by the other clients. (b) Detailed architecture with the same client-side computation boxed separately from the FedAvg aggregation server.

each image label, at the class level, with a synthetic clinical-style note assembled from observation, symptom, context, and indicator slots. To keep the network from memorizing an obvious keyword we engineered the mix deliberately: 25% of notes draw every slot from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This cuts direct label wording, but image and note do not come from the same patient and must never be described as a real image–report pair.

The corpus supports internal comparison, not absolute claims. Every configuration (one choice of aggregation rule, fusion operator, initialization, and anti-collapse components) trains and is scored on the same corpus, so a gap between two of them is attributable to the factor under test rather than to the data. Absolute scores carry no such guarantee. Templates can leave shortcuts, label-level pairing strips out the discordance and missingness of real records, each class is drawn from a different source collection, and the split is grouped by neither patient nor source.

D. Experimental Setup

Hyperparameters. Table I collects the protocol. Every client trains with AdamW (learning rate 10^{-4} , weight decay 0.01) for three local epochs per round. The suite runs eight rounds at batch size 16 in FP32, without automatic mixed precision (AMP). Our reference setting is five clients at Dirichlet $\alpha = 1.0$, and Section IV varies α and the client count explicitly. Seeds 0 and 1 drive initialization, sampling, and partitioning, with one exception: pooled-start reuses a single seed-0 checkpoint under seed-specific FL partitions. We ran the frozen v3 suite in PyTorch 2.13.0+cu130 on one NVIDIA H100 NVL device; the complete suite took 6.46 hours. The GPU-synchronized timer wraps sequential local training and aggregation within a round, and excludes validation, setup, I/O, networking, concurrency, and multi-node throughput. Our timings are an execution-cost record, not distributed throughput. We also did not force deterministic CUDA algorithms, so a seed fixes the software pseudorandom streams without buying identical reruns.

Partitioning. For each class, a Dirichlet draw allocates training indices across clients. Smaller α concentrates client label distributions, and $\alpha \rightarrow \infty$ approaches independent and identically distributed (IID) allocation. A nominal α does not fully describe a finite split, so we report the per-client class shares we actually drew alongside our scalability results. The split is fixed across rounds for a given seed. Shards holding fewer than four examples skip local optimization but still contribute the unchanged global state at their sample weight. We do not renormalize over active clients, which is why our cells report active alongside nominal K .

Matched recent-method controls. Our FedMME-style baseline uploads each client model once and applies the one-shot voting structure in [15]. Our implementation uses equal-weight hard voting and breaks ties by mean softmax. We run it at two local budgets. Our matched 24 epochs isolates the method from the budget; FedMME’s native 100 epochs removes the budget discrepancy. Reporting both keeps the two confounds separate. Either way it keeps the common corpus, shard profile, encoders, and optimizer, substituting our five-class inputs for FedMME’s generated-report pipeline. The P-FIN-style stress test fixes three multimodal and two image-only clients, projects both encoders to 256-dimensional normalized features, and compares zero filling, deterministic FIN, Gaussian β -negative-log-likelihood (β -NLL) imputation ($\beta=0.5$), and uncertainty-weighted aggregation with Fed-UQ-Avg balance weight 0.6 and temperature 0.2 [16]. We kept concatenation instead of P-FIN’s bidirectional cross-modal attention: with one pooled feature per modality, cross-attention carries unit weight and collapses to a linear projection. FedProx uses $\mu=0.01$. Each ablation changes one component and leaves every other setting at the shared defaults that follow, most of which are fixed by what the hardware allows. Three local epochs across eight rounds give a tractable 24-local-epoch budget, and batch size 16 is what holds both encoders plus one local model copy in GPU memory for the largest FP32 model. The grid crosses $K=\{3, 5, 10, 20\}$ with $\alpha=\{0.1, 1, 5\}$ and the $K=5$ sweep adds $\alpha=0.3$ and 0.5, so the study spans severe to near-IID skew; $K=5$ is our small cross-silo reference and $K=3-20$ probes count sensitivity. Two

TABLE I
COMMON SIMULATION PARAMETERS.

Parameter	Value	Parameter	Value
Local Epochs (E)	3	Seeds	0, 1
Fed. Rounds (T)	8	Grid K (ref.)	3, 5, 10, 20 (5)
Batch Size	16	Grid α (ref.)	0.1, 1, 5 (1)
AdamW LR	10^{-4}	$K=5$ extra α	0.3, 0.5
Weight Decay	0.01	Precision	FP32, no AMP
Diversity wt. (λ)	1.0	Init.	Public enc. + rand. heads

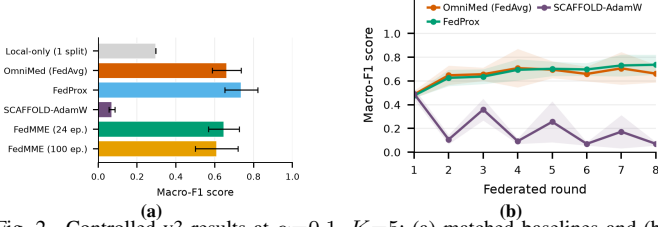


Fig. 2. Controlled-v3 results at $\alpha=0.1$, $K=5$: (a) matched baselines and (b) federated convergence.

seeds support descriptive claims, not significance. Everything else in Table I is a fixed default, not a per-model optimum.

IV. RESULTS

All quantitative results in this section come from the frozen `controlled_v3_real5_template_text` run and its provenance-checked merge. Scores are validation macro-F1. Error bars are sample standard deviations over seeds 0 and 1 unless a single-run diagnostic is identified.

A. Matched Baselines under Label Skew

At $\alpha=0.1$ and $K=5$, local-only training averages 0.297 across the five clients of one partition. Under the same corpus, partition, encoders, optimizer, and 24-local-epoch budget, FedAvg reaches 0.662 ± 0.074 and FedProx 0.737 ± 0.085 (Fig. 2). Their 0.075 gap lies inside the wider sample standard deviation, so two seeds do not separate them. At $\alpha=1$, the same ordering is 0.523 local-only, 0.820 ± 0.029 FedAvg, and 0.862 ± 0.019 FedProx. Thus, both iterative federated methods sit well above isolated client training at both skews.

Our SCAFFOLD-AdamW adaptation reaches only 0.070 ± 0.015 at $\alpha=0.1$ and 0.236 ± 0.205 at $\alpha=1$. Its oscillatory trajectory points to an unstable AdamW control-variate adaptation; this row must not be read as a faithful evaluation of classical SGD-based SCAFFOLD. The FedMME-style one-shot ensemble reaches 0.647 ± 0.080 at the matched 24-epoch budget and 0.609 ± 0.109 at its native 100-epoch budget. Increasing local computation does not repair one-shot aggregation under severe skew.

B. Anti-Collapse, Fusion, and Initialization

The anti-collapse ablation reverses the intended low- α ordering. With the balanced sampler and entropy-diversity term, F1 is 0.683 ± 0.089 . Removing the sampler, the diversity term, or both raises the mean monotonically to 0.716 ± 0.055 , 0.726 ± 0.035 , and 0.758 ± 0.016 (Fig. 3(a)). Every one of the eight $\alpha=0.1$ runs finishes at five-class diversity 1.0, including both unregularized seeds, so final collapse is not observed. The full stack improves the mean minimum diversity across

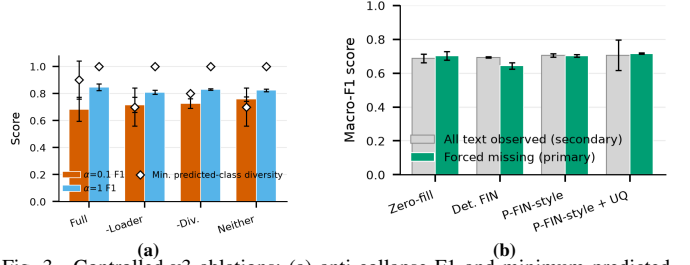


Fig. 3. Controlled-v3 ablations: (a) anti-collapse F1 and minimum predicted-class diversity; (b) forced-missing-text performance.

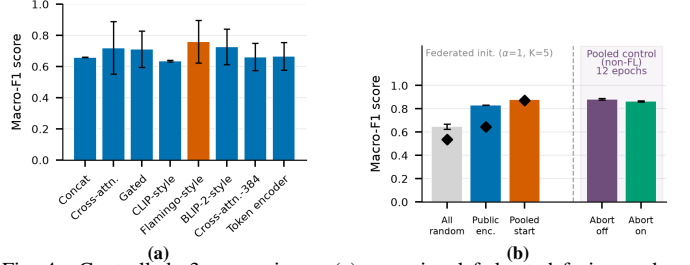


Fig. 4. Controlled-v3 comparisons: (a) operational federated fusion under severe skew; (b) federated initialization and non-federated abort controls.

rounds from 0.70 to 0.90, but it does so with a 0.075 F1 deficit to the unregularized configuration. At $\alpha=1$, all four configurations maintain diversity 1.0 and score 0.809–0.846. We therefore treat the stack as a transient coverage guard, not an accuracy-improving component. Likewise, the non-federated early-abort controls score 0.899 ± 0.022 with abort disabled and 0.871 ± 0.012 with it enabled; neither branch collapses.

The pooled-data fusion screen is a diagnostic architecture comparison, not an operational FL result. Across two seeds, its eight operators span 0.750–0.925 F1. The 0.1755 spread is 33 times the 0.0053 median seed standard deviation, so fusion choice carries real signal in this screen. Residual cross-attention leads at 0.925 ± 0.001 , while 384-dimensional cross-attention is last at 0.750 ± 0.002 ; with only two seeds, the closely grouped leading methods remain descriptively rather than statistically ordered. In the separate operational sweep at $\alpha=0.1$, means span 0.636–0.759 but one operator’s sample standard deviation reaches 0.167, so that severe-skew sweep does not resolve an operational winner (Fig. 4(a)).

The initialization control in Fig. 4(b) closes at 0.647 ± 0.022 with all weights random and 0.831 ± 0.001 with public encoders and random task heads. A pooled-data checkpoint opens at 0.874 and closes at 0.877 ± 0.007 , but it was selected on the reporting split and is a non-deployable diagnostic oracle. The operational public-encoder initialization retains 94.7% of its final F1 without accessing pooled training data.

Under the forced-missing-text metric in Fig. 3(b), zero filling, deterministic FIN, Gaussian β -NLL imputation, and uncertainty-weighted aggregation score 0.688 ± 0.025 , 0.693 ± 0.004 , 0.706 ± 0.010 , and 0.706 ± 0.089 . The uncertainty-weighted rule ties the plain probabilistic rule and is the noisiest, so this matched adaptation does not show that uncertainty weighting tightens performance.

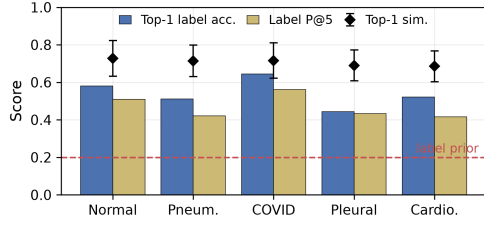


Fig. 5. Controlled-v3 retrieval metrics by condition over 600 held-out queries and a 2,400-note index.

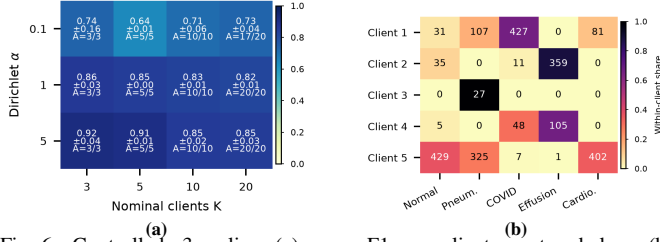


Fig. 6. Controlled-v3 scaling: (a) macro-F1 over client count and skew; (b) one severe-skew client allocation.

C. Retrieval-Stage Evaluation

Exact inner-product FAISS search [22] over training-fitted TF-IDF note vectors gives condition-macro averages of 0.542 top-1 same-label accuracy, 0.470 precision@5, and 0.706 mean top-1 cosine similarity across all 600 validation queries (Fig. 5). Top-1 accuracy is above the 0.2 label prior but well below the similarity score, consistent with synthetic templates whose neighbors can be lexically close without sharing a label. This is a retrieval-stage diagnostic and establishes no clinical generation quality.

D. Scalability, Communication, and Resource Cost

Figure 6(a) crosses $K=\{3, 5, 10, 20\}$ with $\alpha=\{0.1, 1, 5\}$. Every $\alpha=5$ cell lies in 0.849–0.916 and every $\alpha=1$ cell in 0.818–0.862, whereas $\alpha=0.1$ yields 0.643–0.738 with spreads up to ± 0.159 . Within a row, increasing K nearly sevenfold moves the mean by at most 0.10 F1, while changing skew moves it by as much as 0.27. Client count is therefore secondary to realized label allocation in this grid. Cells show A/K , where A counts shards with at least four samples; the severe $K=20$ split has 17 active clients. Figure 6(b) shows one realized partition in which individual clients hold nearly entire classes.

The GPU-synchronized sections in Fig. 7 take 41–62 s per round and allocate 4.77–5.92 GiB on the shared H100. These panels are execution records, not distributed-throughput curves: clients are simulated sequentially, and uncontrolled server contention remains.

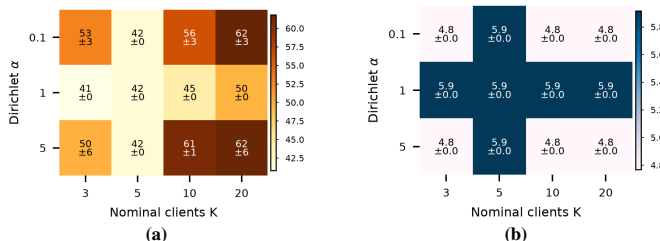


Fig. 7. Controlled-v3 resource measurements: (a) seconds per round and (b) peak allocated GPU memory.

For $|\theta|$ trainable parameters, b bytes per parameter, T rounds, and nominal K , the plotted bidirectional volume is

$$V_{\text{nom}} = 2KT|\theta|b. \quad (3)$$

Here $|\theta|=153,935,621$, $b=4$, and $T=8$, giving 27.5/45.9/91.8/183.5 GiB at $K=3/5/10/20$. These are model-state calculations, not measured network traffic; serialization, compression, secure aggregation [23], and algorithm-specific state are excluded.

The matched single-seed branch audit at $\alpha=1$, $K=5$ scores text, image, and concatenated multimodal models at 0.880, 0.768, and 0.830 (Fig. 8(b)). Thus, concatenation improves on image-only by 0.062 but trails text-only by 0.050 in this run. Its one-way FP32 state is 0.573 GiB versus 0.248/0.324 GiB, total timed rounds take 5.6 versus 1.8/4.3 minutes, and peak allocation is 5.92 versus 2.84/4.33 GiB. Fusion therefore incurs a clear systems cost without guaranteeing a gain over the stronger unimodal branch.

V. DISCUSSION AND LIMITATIONS

The v3 rerun supports the paper’s systems thesis but narrows two model claims. Federation provides a large advantage over isolated local training, and partition skew dominates client count. Fusion architecture also matters in the pooled diagnostic screen. However, the operational concatenated model does not beat text-only in the matched branch audit, and the anti-collapse stack does not improve F1 under severe skew. We retain both negative findings rather than overstate claims contradicted by v3.

The corpus remains a proxy: notes are synthetic and class-paired, not patient-paired; template wording can leak class information; image class is partly confounded with source collection; and the split is grouped by neither patient nor source. Two seeds permit descriptive spreads, not significance, and deterministic CUDA algorithms were not forced. Sequential single-GPU simulation omits concurrency, latency, stragglers, dropout, and heterogeneous hardware. Communication is analytic model-state volume rather than measured network traffic. FedMME, P-FIN, and SCAFFOLD rows are declared adaptations, not native replications. Consequently, none of the F1 or retrieval values estimates clinical safety, generalization, or deployment readiness.

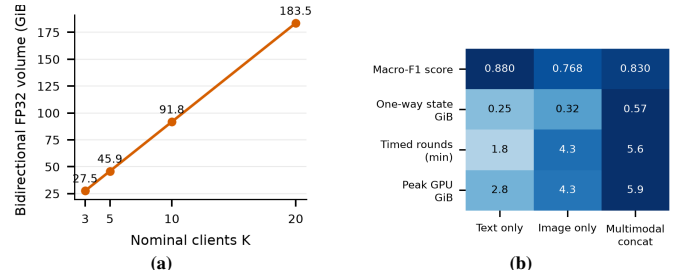


Fig. 8. Controlled-v3 systems cost: (a) formula-derived FP32 model-state volume; (b) single-seed modality score and measured resource audit.

VI. CONCLUSION

On 3,000 public radiographs with class-paired synthetic notes, matched controlled-v3 experiments show that FedAvg and FedProx outperform local-only training at both tested skews, while label skew changes F1 more than a near-sevenfold change in client count. Public encoders retain 94.7% of a pooled diagnostic oracle without pooled operational training. The fusion screen shows genuine architecture sensitivity, but concatenation trails text-only in the matched federated branch audit and costs $2.3\times$ its model state. Removing both anti-collapse components improves severe-skew F1 from 0.683 to 0.758, with all runs finishing at full predicted-class diversity. These findings define OmniMed-FL as a controlled systems benchmark and motivate patient-paired, multi-institutional validation with more seeds, native baseline implementations, and end-to-end network measurements.

ACKNOWLEDGMENT

This work was supported in part by the Indian National Academy of Engineering (INAE) Chair Professorship grant awarded to the third author.

REFERENCES

- [1] J. Irvin, P. Rajpurkar, M. Ko, *et al.*, “CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 2019, pages 590–597.
- [2] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the International Conference on Artificial Intelligence and Statistics*, volume 54, 2017, pages 1273–1282.
- [3] M. J. Sheller, B. Edwards, G. A. Reina, *et al.*, “Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data,” *Scientific Reports*, volume 10, article 12598, 2020.
- [4] Q. Dou, T. Y. So, M. Jiang, *et al.*, “Federated deep learning for detecting COVID-19 lung abnormalities in CT: A privacy-preserving multinational validation study,” *npj Digital Medicine*, volume 4, article 60, 2021.
- [5] P. Rajpurkar, J. Irvin, K. Zhu, *et al.*, “CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning,” *arXiv:1711.05225*, 2017.
- [6] A. Dosovitskiy, L. Beyer, A. Kolesnikov, *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proceedings of the International Conference on Learning Representations*, 2021.
- [7] L. Lin, Y. Liu, J. Wu, *et al.*, “FedLPPA: Learning personalized prompt and aggregation for federated weakly-supervised medical image segmentation,” *IEEE Transactions on Medical Imaging*, volume 44, number 3, pages 1127–1139, 2025.
- [8] D. Chu, W. Jaafar, and H. Yanikomeroglu, “On the design of communication-efficient federated learning for health monitoring,” in *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, 2022, pages 1128–1133.
- [9] T. Li, A. K. Sahu, M. Zaheer, *et al.*, “Federated optimization in heterogeneous networks,” in *Proceedings of Machine Learning and Systems*, volume 2, 2020, pages 429–450.
- [10] S. P. Karimireddy, S. Kale, M. Mohri, *et al.*, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proceedings of the International Conference on Machine Learning*, volume 119, 2020, pages 5132–5143.
- [11] Y. Tian, Z. Zhang, Z. Yang, and R. Jin, “Hierarchical federated learning with adaptive clustering on non-IID data,” in *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, 2022, pages 627–632.
- [12] S. Zhang, Y. Xu, N. Usuyama, *et al.*, “A multimodal biomedical foundation model trained from fifteen million image–text pairs,” *NEJM AI*, volume 2, number 1, article A10a2400640, 2025.
- [13] C. Li, C. Wong, S. Zhang, *et al.*, “LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day,” in *Proceedings of Advances in Neural Information Processing Systems*, volume 36, 2023, pages 28541–28564.
- [14] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, “MedCLIP: Contrastive learning from unpaired medical images and text,” in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2022, pages 3876–3887.
- [15] N. Wang, Y. Deng, S. Fan, *et al.*, “Multi-modal one-shot federated ensemble learning for medical data with vision large language model,” *arXiv:2501.03292*, 2025.
- [16] N. F. Shahid, M. Ahmed, M. A. Haider, *et al.*, “Probabilistic feature imputation and uncertainty-aware multimodal federated aggregation,” in *Proceedings of Medical Imaging with Deep Learning*, volume 315, 2026, pages 3593–3607.
- [17] Y. Yan, H. Wang, Y. Huang, *et al.*, “Cross-modal vertical federated learning for MRI reconstruction,” *IEEE Journal of Biomedical and Health Informatics*, volume 28, number 11, pages 6384–6394, 2024.
- [18] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” *arXiv:1910.01108*, 2019.
- [19] D. S. Kermany, M. Goldbaum, W. Cai, *et al.*, “Identifying medical diagnoses and treatable diseases by image-based deep learning,” *Cell*, volume 172, number 5, pages 1122–1131.e9, 2018.
- [20] X. Wang, Y. Peng, L. Lu, *et al.*, “ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pages 2097–2106.
- [21] M. E. H. Chowdhury, T. Rahman, A. Khandakar, *et al.*, “Can AI help in screening viral and COVID-19 pneumonia?” *IEEE Access*, volume 8, pages 132665–132676, 2020.
- [22] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, volume 7, number 3, pages 535–547, 2021.
- [23] I. Ergun, H. U. Sami, and B. Güler, “Communication-efficient secure aggregation for federated learning,” in *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, 2022, pages 3881–3886.
- [24] P. H. Barros and H. S. Ramos, “A novel aggregation method to promote safety security for poisoning attacks in federated learning,” in *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, 2022, pages 3869–3874.